

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Circles

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02

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**A**

A. The center is at $(13, k)$ and the radius is 8.

Choice A is correct. For a circle in the xy -plane that has the equation $x - h^2 + y - k^2 = r^2$, where h , k , and r are constants, h, k is the center of the circle and the positive value of r is the radius of the circle. In the given equation, $h = 13$ and $r^2 = 64$. Taking the square root of each side of $r^2 = 64$ yields $r = \pm 8$. Therefore, the center of the circle is at $13, k$ and the radius is 8.

Choice B is incorrect. This gives the center and radius of a circle with equation $x - k^2 + y - 13^2 = 64$, not $x - 13^2 + y - k^2 = 64$.

Choice C is incorrect. This gives the center and radius of a circle with equation $x - k^2 + y - 13^2 = 4,096$, not $x - 13^2 + y - k^2 = 64$.

Choice D is incorrect. This gives the center and radius of a circle with equation $x - 13^2 + y - k^2 = 4,096$, not $x - 13^2 + y - k^2 = 64$.

Q2**5**

The correct answer is 5. It's given that in the xy -plane, the graph of the equation $(x - 3)^2 + (y - 5)^2 = 9$ is a circle. It's also given that the point $(6, c)$, where c is a constant, lies on this circle. It follows that the ordered pair $(6, c)$ makes the equation $(x - 3)^2 + (y - 5)^2 = 9$ true. Substituting 6 for x and c for y in this equation yields $(6 - 3)^2 + (c - 5)^2 = 9$, or $9 + (c - 5)^2 = 9$. Subtracting 9 from each side of this equation yields $(c - 5)^2 = 0$. It follows that the value of c is 5.

Q3**D****D.** $-29, 0$

Choice D is correct. It's given that in the xy -plane, the graph of $x^2 + 58x + y^2 = 0$ is a circle. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the coordinates of the center of the circle are (h, k) and the radius of the circle is r . By completing the square, the equation $x^2 + 58x + y^2 = 0$ can be rewritten as $x^2 + 58x + \frac{58^2}{2} + y^2 = 0 + \frac{58^2}{2}$, or $x^2 + 58x + 841 + y^2 = 841$. This equation is equivalent to $(x + 29)^2 + y^2 = 841$, or $(x - (-29))^2 + (y - 0)^2 = 841$. Therefore, h is -29 and k is 0 , and the coordinates (x, y) of the center of the circle are $(-29, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4**C****C.** 195

Choice C is correct. It's given that the measure of angle R is $\frac{2\pi}{3}$ radians, and the measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R . Therefore, the measure of angle T is equal to $\frac{2\pi}{3} + \frac{5\pi}{12}$ radians. Multiplying $\frac{2\pi}{3}$ by $\frac{4}{4}$ to get a common denominator with $\frac{5\pi}{12}$ yields $\frac{8\pi}{12}$. Therefore, $\frac{2\pi}{3} + \frac{5\pi}{12}$ is equivalent to $\frac{8\pi}{12} + \frac{5\pi}{12}$, or $\frac{13\pi}{12}$. Therefore, the measure of angle T is $\frac{13\pi}{12}$ radians. The measure of angle T , in degrees, can be found by multiplying its measure, in radians, by $\frac{180}{\pi}$. This yields $\frac{13\pi}{12} \times \frac{180}{\pi}$, which is equivalent to 195 degrees. Therefore, the measure of angle T is 195 degrees.

Choice A is incorrect. This is the number of degrees that the measure of angle T is greater than the measure of angle R .

Choice B is incorrect. This is the measure of angle R , in degrees.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**C****C.** 1, - 7

Choice C is correct. The equation of a circle in the xy -plane can be written as

$(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that the circle in the xy -plane is defined by the equation $(x - 1)^2 + (y + 7)^2 = 1$. This equation can be written as $(x - 1)^2 + (y - (-7))^2 = 1$. For this equation, it follows that $h = 1$ and $k = -7$. Therefore, the center of the circle in the xy -plane defined by the given equation is $(1, -7)$.

Choice A is incorrect. This is the center of the circle in the xy -plane that is defined by the equation $(x + 1)^2 + (y + 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

Choice B is incorrect. This is the center of the circle in the xy -plane that is defined by the equation $(x + 1)^2 + (y - 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

Choice D is incorrect. This is the center of the circle in the xy -plane that is defined by the equation $(x - 1)^2 + (y - 7)^2 = 1$, not $(x - 1)^2 + (y + 7)^2 = 1$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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